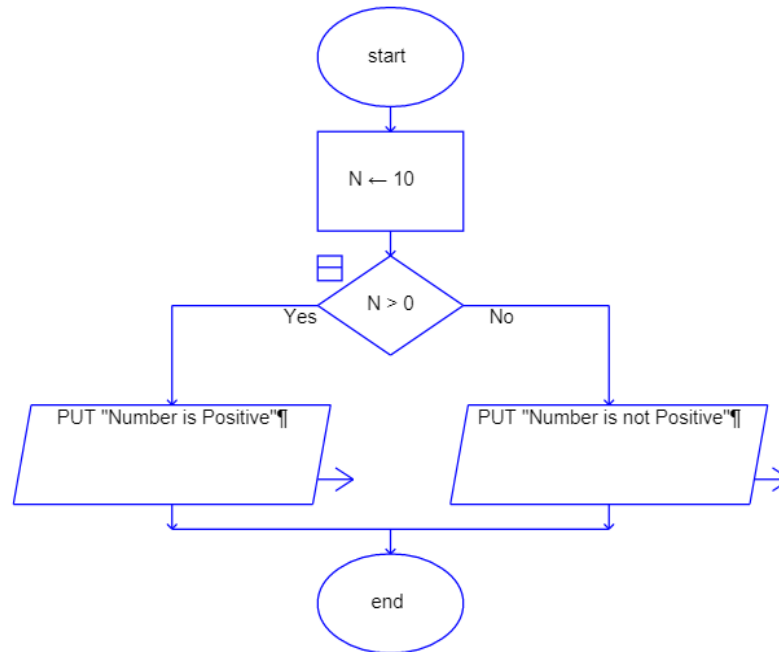


Assignment No.1

1. Check Positive Number:

- Create a flowchart to check whether a number is positive.



- Write a Java program that checks if a predefined number is positive using an if-else statement and prints the appropriate message.

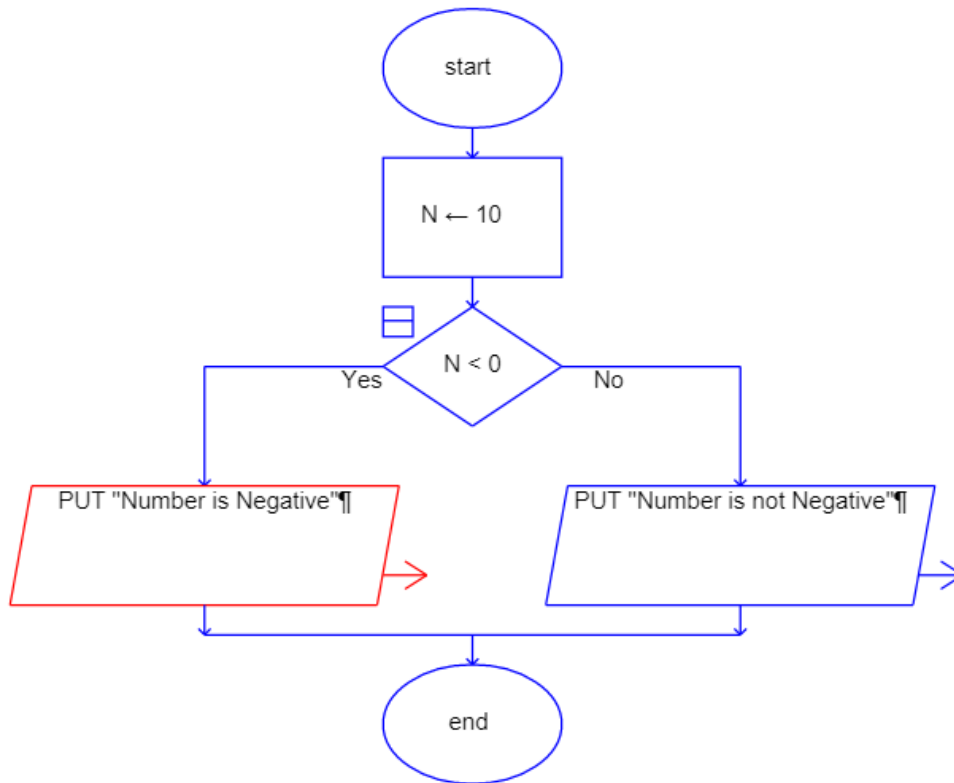
Code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int n = 10;
        if (n > 0) {
            System.out.println("Number is Positive");
        } else {
            System.out.println("Number is not Positive");
        }
    }
}
```

Output: Number is Positive

2. Check Negative Number:

- Create a flowchart to check whether a number is negative.



- Write a Java program that checks if a predefined number is negative using an if-else statement and displays the result.

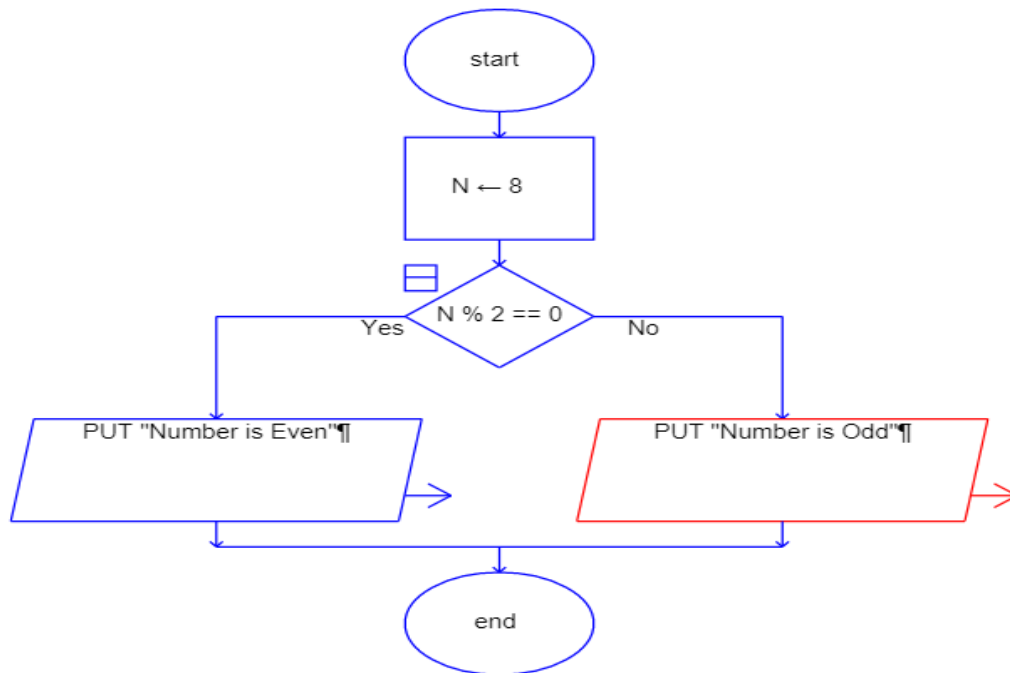
Code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int n = 10;
        if (n < 0) {
            System.out.println("Number is Negative");
        } else {
            System.out.println("Number is not Negative");
        }
    }
}
```

Output: **Number is Negative**

3. Check Odd or Even Number:

- Create a flowchart to determine whether a number is odd or even.



- Write a Java program that checks if a predefined number is odd or even. Use an if-else statement and the modulus operator (%) to determine whether the number is divisible by 2 or not.

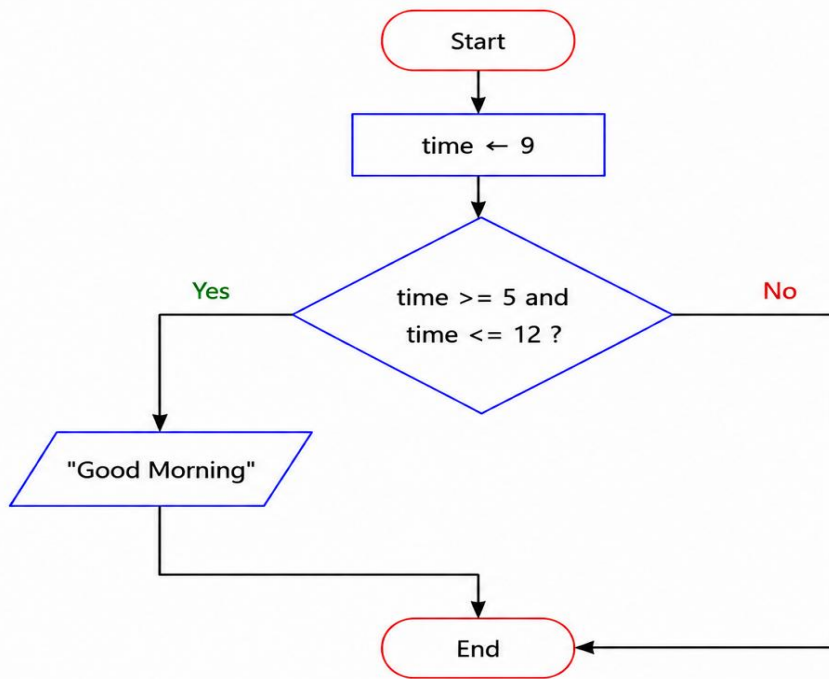
Code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int n = 8;
        if (n % 2 == 0) {
            System.out.println("Number is Even");
        } else {
            System.out.println("Number is Odd");
        }
    }
}
```

Ouput: **Number is Even**

4. Display Good Morning Message Based on Time:

- Create a flowchart to display a "Good Morning" message based on a given time.



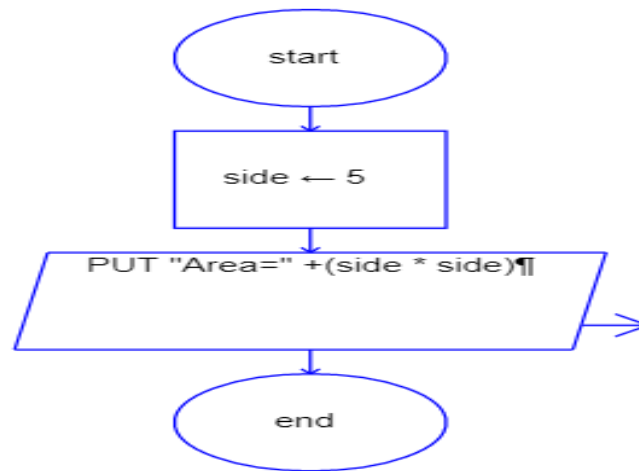
- Write a Java program that displays a "Good Morning" message if the predefined time is between 5 AM and 12 PM. Use an if statement to implement the logic.
Code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int time = 9;
        if (time >= 5 && time <= 12) {
            System.out.println("Good Morning");
        }
    }
}
```

Output: **Good Morning**

5. Print Area of a Square:

- Create a flowchart to calculate and print the area of a square.



- Write a Java program that calculates the area of a square using the formula $\text{area} = \text{side} * \text{side}$. Use a predefined side length.

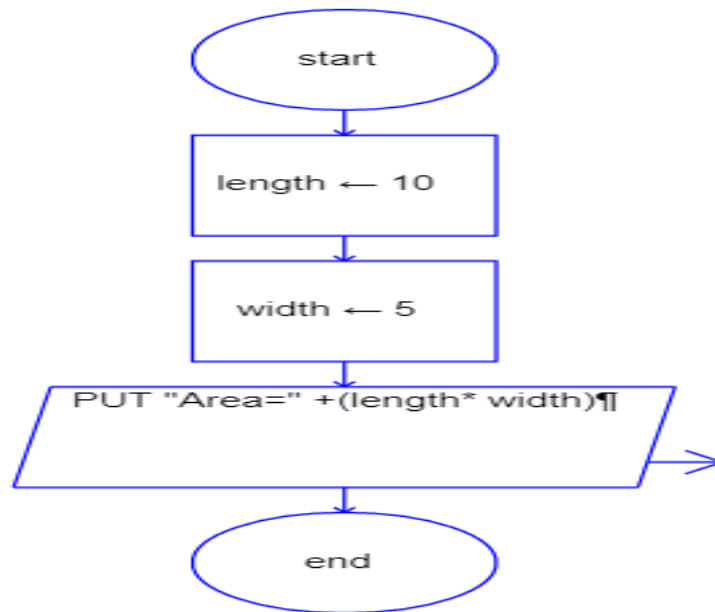
code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int side = 5;
        int area = side * side;
        System.out.println("Area of square = " + area);
    }
}
```

Output: **Area of square = 25**

6. Print Area of a Rectangle:

- Create a flowchart to calculate and print the area of a rectangle.



- Write a Java program that calculates the area of a rectangle using the formula $\text{area} = \text{length} * \text{width}$. Use predefined values for length and width.

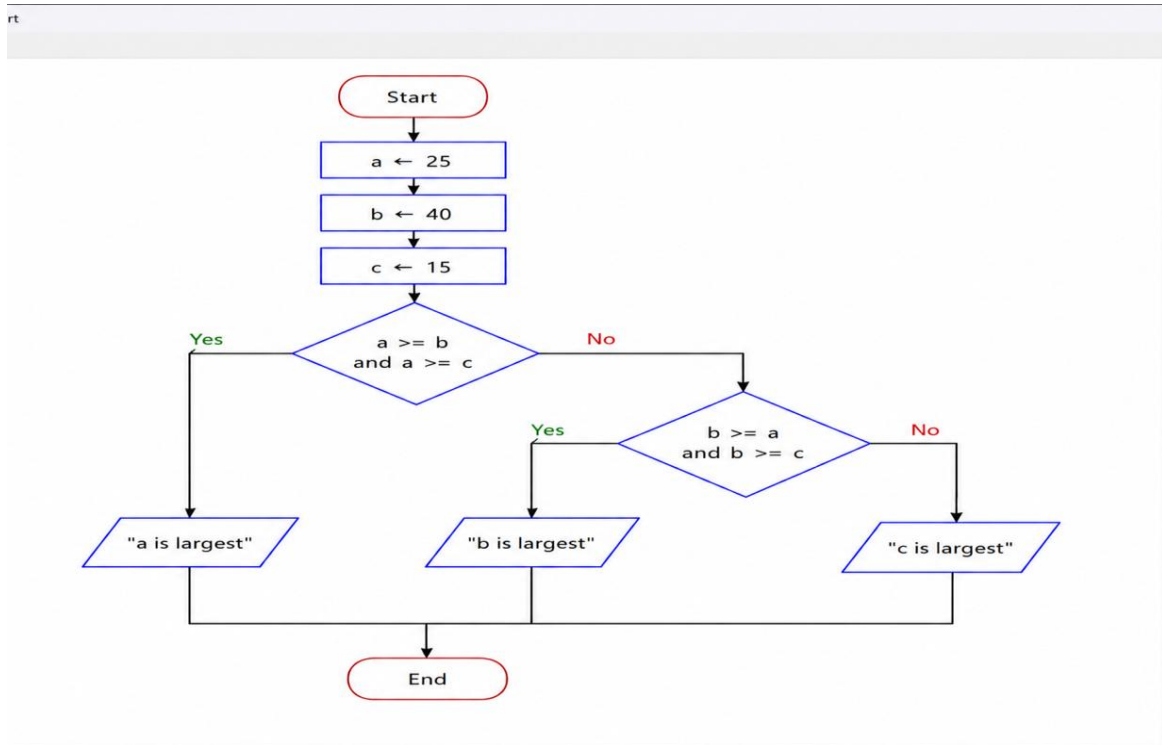
code:

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        int length = 10;
        int width = 5;
        int area = length * width;
        System.out.println("Area of rectangle = " + area);
    }
}
```

Output: **Area of rectangle = 50**

7. Find the Largest of Three Numbers:

- Create a flowchart to find the largest of three numbers.



- Write a Java program that finds and prints the largest of three predefined numbers using if-else statements.

Code:

```
public class Main {  
    public static void main(String[] args) {  
        int a = 25;  
        int b = 40;  
        int c = 15;  
        if (a >= b && a >= c) {  
            System.out.println("Largest number = " + a);  
        } else if (b >= a && b >= c) {  
            System.out.println("Largest number = " + b);  
        } else {  
            System.out.println("Largest number = " + c);  
        }  
    }  
}
```

Output: **Largest number = 40**